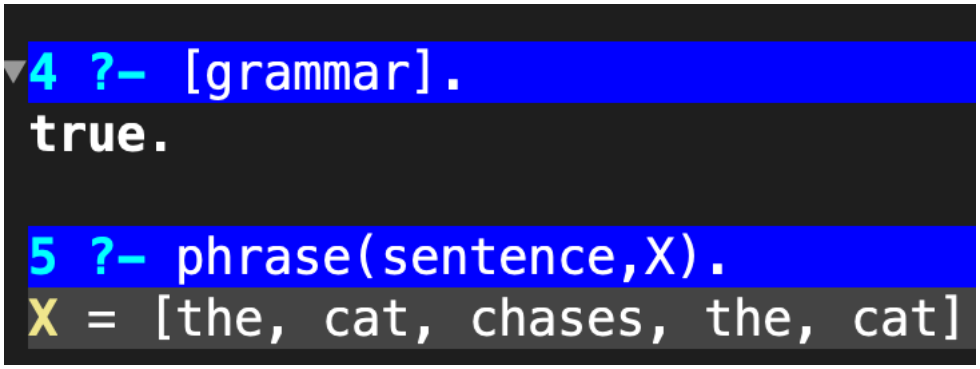
LAB 09 ANSWERS REPORT

HANS PUJALTE (PUJA0009)

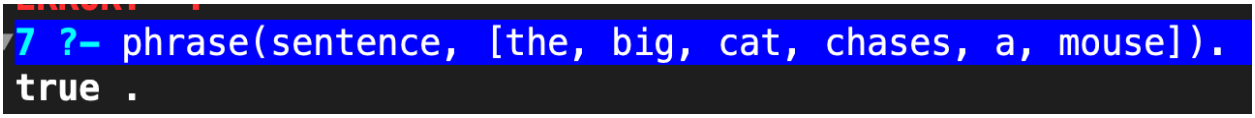
COMP3742, TONSLEY
COMPLETED: 26/10/2026

Hans Pujalte

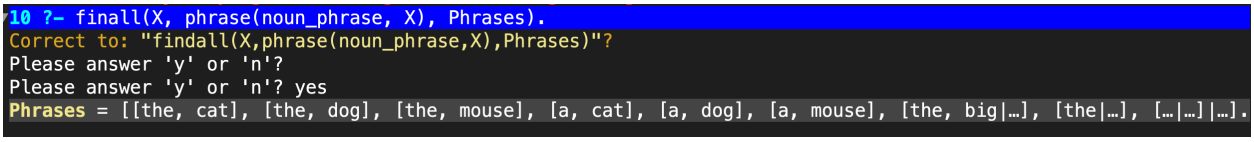
1. Generate a sentence using the grammar.pl knowledge base using Prolog.



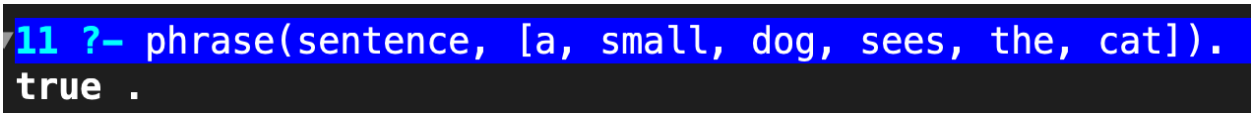
2. Parse the sentence “the big cat chases a mouse” using the grammar.pl knowledge base and Prolog.



3. List all possible noun phrases that can be generated using the context-free grammar specified in the grammar.pl knowledge base.



4. Check to see if “a small dog sees the cat” is a valid sentence according the context-free grammar described in grammar.pl .



5. Extend the context-free grammar described in grammar.pl to include prepositional phrases. Once the grammar is extended use it generate a sentence with a prepositional phrase.

```
3   % Sentence structure
4   sentence --> noun_phrase, verb_phrase, preposition_phrase.
5
6   % Noun phrase structure
7   noun_phrase --> determiner, noun.
8   noun_phrase --> determiner, adjective, noun.
9
10  % Verb phrase structure
11  verb_phrase --> verb, noun_phrase.
12  verb_phrase --> verb.
13
14  % preposition phrase structure
15  preposition_phrase --> preposition, modifier, object.
16  preposition_phrase --> preposition, object.
17
18  % preposition
19  preposition --> [on].
20  preposition --> [in].
21  preposition --> [with].
22  preposition --> [under].
23
24  % Modifier
25  modifier --> [the].
26  modifier --> [my].
27
28  % Object
29  object --> [table].
30  object --> [bed].
31  object --> [couch].
```

```
18 ?- phrase(sentence, [a, small, dog, sees, the, cat, under, the, bed]).
true
```